

Permutation Representation

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Problem 1. Let G act on the cosets G/H , by left multiplication. Show that the kernel (ie the kernel of the map $G \rightarrow \text{Perm}(G/H)$) is the intersection of all conjugates of H . Show that this is a normal subgroup of G , contained in H , and moreover it contains every normal subgroup of G that is contained in H , so its the unique largest subgroup of H which is normal in G . Call this subgroup $LN(H)$ (for largest normal in H , this is not at all standard notation).

Claim 1a. Show that the kernel (ie the kernel of the map $G \rightarrow \text{Perm}(G/H)$) is the intersection of all conjugates of H .

Proof. By the previous homework we know the stablizer of $aH \in G/H$ is aHa^{-1} . The elements of the kernel stablize all cosets as they map each coset to the identity thus it has to be intersections of the stablizer of all the cosets and thus $\bigcap_{g \in G} gHg^{-1}$ $\square$

Claim 1b. Show that this is a normal subgroup of G

Proof. $\forall h \in G$ and $h(\bigcap_{g \in G} gHg^{-1})h^{-1} = \bigcap_{g \in G} hgHg^{-1}h^{-1} = \bigcap_{g \in G} gHg^{-1}$. Thus $LN(H)$ is normal. Also kernel of homomorphism so is a normal subgroup. $\square$

Claim 1c. $\bigcap_{g \in G} gHg^{-1} \subseteq H$

Proof. $\forall h \in \bigcap_{g \in G} gHg^{-1}$, $h \in eHe = H$. Thus $\bigcap_{g \in G} gHg^{-1} \subseteq H$. $\square$

Claim 1d. $\forall N \triangleleft G \ni N \subseteq H, N \subseteq \bigcap_{g \in G} gHg^{-1}$

Proof. Suppose $N \triangleleft G \ni N \subseteq H$. $\forall g \in G$, $gNg^{-1} = N$ and since $N \subseteq H$, $\forall g \in G$, $gNg^{-1} = N \subseteq gHg^{-1}$. Thus $N \subseteq \bigcap_{g \in G} gHg^{-1}$ $\square$

Problem 2. Let Q be the quaternion group. Show that every non-trivial subgroup contains $Z = \{1, -1\}$, the center of Q . Conclude that the only way that $G = Q$ can act faithfully on the disjoint union of a bunch of G/H_i is if one of the H_i is trivial. Conclude that Q cannot act faithfully on a set with less than 8 elements. Which set with 8 does it act faithfully on?

Claim 2a. Show that every non-trivial subgroup contains $Z = \{1, -1\}$, the center of Q .

Proof. $\forall S \subseteq Q$ where S is a non trivial subgroup. $1 \in S$ because 1 is identity of Q . If $S = Z$, $Z \subseteq S$. If $i \in S$, then $i * i = -1 \in S$. Similarly if $j \in S$, $-1 \in S$ and $k \in S$, $-1 \in S$. If $-i \in S$, $-i * -i = i^2 = -1 \in S$. Similarly if $-j \in S$ or $-k \in S$ then $-1 \in S$. All this is by closure. So $Z \subseteq S$. $\square$

Claim 2b. Conclude that the only way that $G = Q$ can act faithfully on the disjoint union of a bunch of G/H_i is if one of the H_i is trivial.

Proof. If G acts faithfully on a set it acts faithfully on a disjoint union of quotient sets. But since $\forall H_i \neq \{e\} \ Z \subset H_i, \forall H_i \neq \{e\} \ Z \subseteq \bigcap_{i \in 1 \dots k} LN(H_i)$ thus G cannot act faithfully if $\bigcup H_i = \{e\}$. Thus G can only act faithfully if $\exists H_i = \{e\}$. Thus $\bigcap_{i \in 1 \dots k} LN(H_i) = e$. $\square$

Claim 2c. $G \subset S_8$

Proof. Thus since $\exists H_i = e$, Q must act faithfully on a set of 8 elements and cannot act faithfully on a set of less than 8 elements because then $H_i \neq e$ and thus contradicting the previous theorem.

Since Q cannot act faithfully on a set with less than 8 elements but acts faithfully on a set of 8 elements it must act on S_8 . $\square$

Problem 3. Show that aHg^{-1} is a coset for gHg^{-1} . Show that $aH \rightarrow aHg^{-1}$ gives a well defined map $G/H \rightarrow G/gHg^{-1}$, and moreover this is an isomorphism of G -sets (the map is a bijection, commuting with the action of G by left mult on both quotients). Show that the actions of G on G/H and G/K are isomorphic iff H and K are conjugate.

Claim 3a. Show that aHg^{-1} is a coset for gHg^{-1} .

Proof. For $gHg^{-1} \in G/gHg^{-1}$. Left multiply by ag^{-1} thus $ag^{-1}gHg^{-1} = aHg^{-1}$. Thus $aHg^{-1} \in G/gHg^{-1}$ $\square$

Claim 3b. Show that $aH \rightarrow aHg^{-1}$ gives a well defined map $\alpha : G/H \rightarrow G/gHg^{-1}$

Proof. Suppose $aH, bH \in G/H \ni aH = bH$. Then $\exists h \in H \ni b = ah$. Thus $bH = ahH$. $aH \rightarrow aHg^{-1}$ and $bH = ahH \rightarrow ahHg^{-1} bHg^{-1} = ahHg^{-1} = aHg^{-1}$. The map is well defined. $\square$

Claim 3c. α is isomorphism of G sets.

Proof. $\forall agHg^{-1} \in G/gHg^{-1}, agHg^{-1} = (a'g^{-1})gHg^{-1} = a'Hg^{-1} \ni a = a'g^{-1} \exists a'H \in G/H$ thus $\alpha(a'H) = a'Hg^{-1}$. Thus α is surjective.

$\forall aH, bH \in G/H \ni \alpha(aH) = \alpha(bH)$ then $aHg^{-1} = bHg^{-1}$ then $\exists h_1, h_2 \in H$ s.t. $ah_1g^{-1} = bh_2g^{-1} \implies b = ah_1h_2^{-1}$. Thus $b \in aH$ and $aH = bH$. thus α is injective.

Thus α is bijection.

$\forall x \in G, x \cdot aH = xaH$ by left multiplication. $\alpha(x \cdot aH) = \alpha(xaH) = xaHg^{-1} = x \cdot aHg^{-1} = x \cdot \alpha(aH)$. Thus α is a iso of G sets. $\square$

Claim 3d. Show that the actions of G on G/H and G/K are isomorphic iff H and K are conjugate.

Proof. $\implies$: Suppose the action of G on G/H and G/K are isomorphic. Then there exists a isomorphism of G -sets $\alpha : G/H \rightarrow G/K$. Thus $\exists X$ some set where the action on X is transitive and is iso to G/H and G/K . Thus $H \cong G^x$ and $K \cong G^y$ for $x, y \in X$. Since the action is transitive, $y = gx$. By (our favorite conjugation formula) previous homework we know two stablizers are conjugates if the elements are in the same orbit. Thus H and K are conjugate.

$\impliedby$: If H and K are conjugate then by the previous parts there exists a isomorphism of G sets. Thus the actions are isomorphic. $\square$

Problem Artin 7a. For D_4 , find the smallest integer n such that the group has a faithful operation on a set of order n :

Solution. Suppose $\phi : D_4 \rightarrow S_n$ is injective. Then $D_4 \subseteq S_n$ in a natural way. Thus $|S_4| \geq |D_4|$. This implies $n \geq 4$ because $|S_2| = 2, |S_3| = 6$ and $|S_4| = 24$. D_4 acts transitively on the $Verts(S)$ the 4 vertices of a square. Because all rotations map any vertex to any other vertex. Suppose $k \in \ker \phi$ then $\forall v \in Verts(S) kv = v$. Thus all 4 vertices are fixed thus there are 4 vector, the vector from 0 to each vertex, which span $\mathbb{R}^2$ which are fixed thus $\mathbb{R}^2$ must be fixed and $k = I$. Thus $\ker \phi$ is trivial and ϕ is injective and faithful. ■

Problem Artin 7b. For D_6 , find the smallest integer n such that the group has a faithful operation on a set of order n :

Solution. Suppose $\phi : D_6 \rightarrow S_n$ is injective. Then $D_6 \subseteq S_n$ in a natural way. $|S_2| = 2$ and $|S_3| = 6$ are invalid choices thus n is 4 or more.

1. S_4 : If D_6 acts on 4 things transitively then $[S_4 : D_6] = 2 \implies D_6 \cong A_4$. But A_4 has no order 6 element.

Cannot act on 1 and 3 items because it just becomes S_3 . Cannot act on 2 and 2 items because $|S_2 \times S_2| = 4 < |D_6|$. Thus $n \neq 4$

2. S_5 : Cannot act transitively on 5 items because $5 \nmid |D_6|$. (2 and 3):

Claim 7bi. $D_6 \cong \mu_2 \times D_3 \cong S_2 \times S_3$

Proof. $D_6 = \langle x, y | x^6 = y^2 = 1, xy = x^{-1}y \rangle$. $D_3 = \langle x^2, y | (x^2)^3 = y^2 = 1, xy = x^{-1}y \rangle$. Thus $D_3 \subseteq D_6$. $\mu_2 = \{1, x^3\}$ where $1 = I$ and x^3 is a rotation by π and thus equal to $-I$. $\mu_2 \subseteq Z(D_6)$ naturally. Thus $\mu_2 D_3 = D_3 \mu_2$. Furthermore $x^3 \in \mu_2$ and $x^4 \in D_3$ we can get $x^3 \times x^4 = x$ and $e \in \mu_2$ and $y \in D_3$ $e \times y = y$. Thus we can get the generators of D_6 thus homomorphism from $\mu_2 \times D_3 \rightarrow D_6$ is surjective. Finally $\{e\} = \mu_2 \cap D_3$. Thus $D_6 \cong \mu_2 \times D_3 \cong S_2 \times S_3$. ■

Thus $D_6 \cong S_2 \times S_3$ which is the same thing as D_6 faithfully permutes 2 and 3 elements separately and thus permutes 5 elements faithfully.

Thus $n = 5$ ■

Problem Artin 7c. For Q_8 , find the smallest integer n such that the group has a faithful operation on a set of order n :

Proof. Already proven in problem 2 □